

SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID** – SIH26055
- **Problem Statement Title**- Smart Scan strategy for Electronic Warfare
- **Theme**- Clean & Green Technology
- **PS Category**- Software
- **Team ID**-
- **Team Name**- PUSHPAK



**SMART INDIA
HACKATHON
2026**

Simulation-only scope: no RF hardware, no interception, no jamming, no weapon control.

Intelligent RF Spectrum Scan Strategy

Detailed explanation of the proposed solution

A machine-learning Electronic Support receiver scheduler — the expected solution, built end to end.

Simulated RF environment carrying truth information for every band at every time slot.

Bandwidth-limited receiver with dwell time, tuning delay and detection threshold.

Bandit trained on hits and misses; periodicity estimator for optimal interception of periodic scans.

How it addresses the problem

Detection rate rises **0.107** → **0.345** on the same spectrum and seeds.

Listening efficiency rises **0.198** → **0.573** of dwell spent on live air.

High-priority detection rises **0.101** → **0.375**.

A decision returns in **6.6 ms p95** over HTTP, against a 50 ms budget.

Innovation and uniqueness of the solution

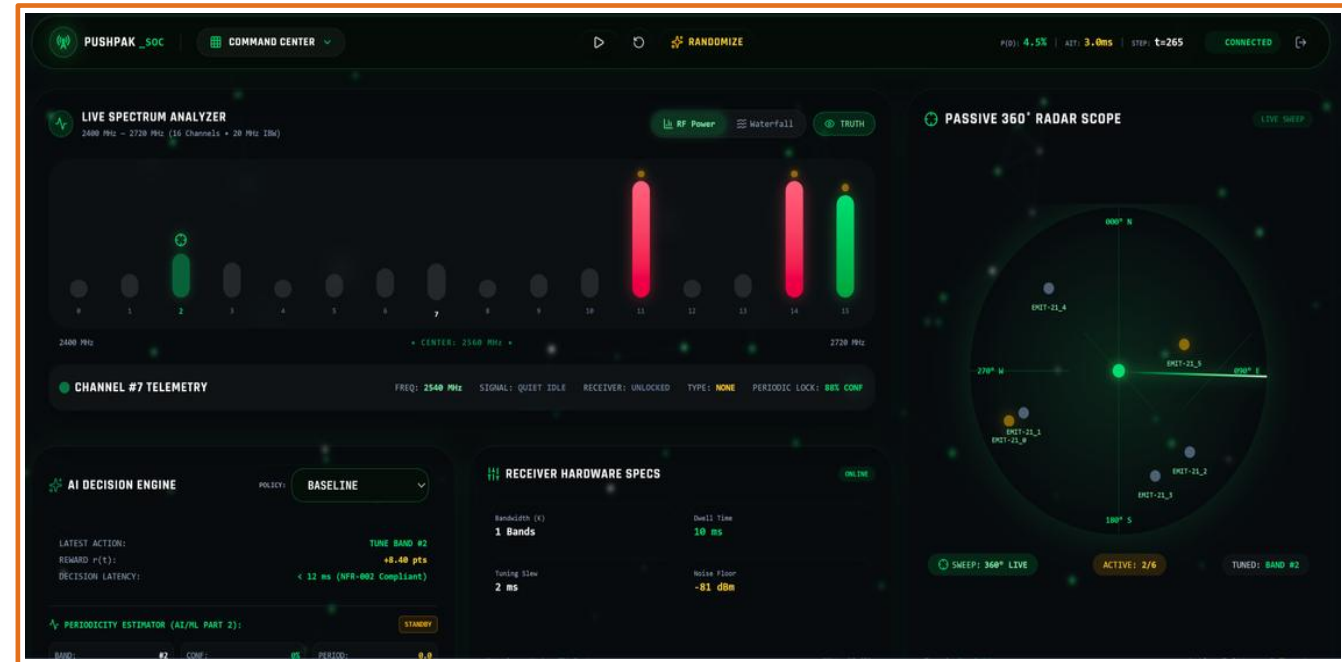
Phase folding, not median inter-arrival — we observe our own detections, never the emitter, so the gaps measure our scan schedule.

Confidence corrected for the look-elsewhere effect — non-periodic emitters stay at the nominal 5%, not 99%.

Coverage versus density is a tunable reward weight, retuned per mission rather than fixed.

3.2×
Detection rate
3.7×
High-priority
2.9×
Scan efficiency

Measured against the round-robin sweep on an identical spectrum — Scenario B, 5 seeds.



Working prototype — live spectrum analyzer, AI decision engine and passive radar scope.

Technologies

Frontend: React 18 + TypeScript — real-time spectrum and metrics dashboard

Backend: Java 21 + Spring Boot 3 — APIs, simulation and scheduler services

AI/ML: Python 3.12 + FastAPI — contextual bandit, Q-learning, DQN/PPO

Database: PostgreSQL + Flyway — simulations, scan events and results

Real-Time: WebSocket — live spectrum, decisions and metrics

Deployment: Docker / Docker Compose — reproducible setup

Implementation

Build RF simulator with configurable bands and 5 emitter behaviours.

Model receiver with bandwidth K, dwell time and tuning delay.

Implement baseline using deterministic round-robin scanning.

Develop ML scheduler using a contextual multi-armed bandit.

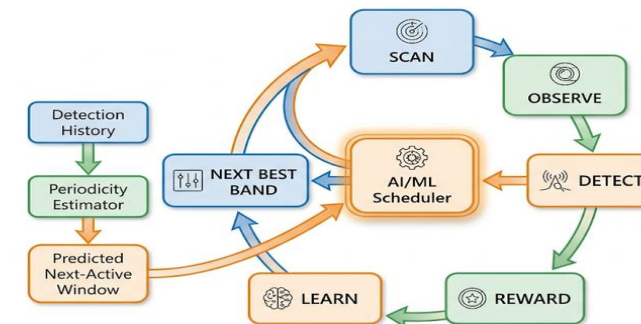
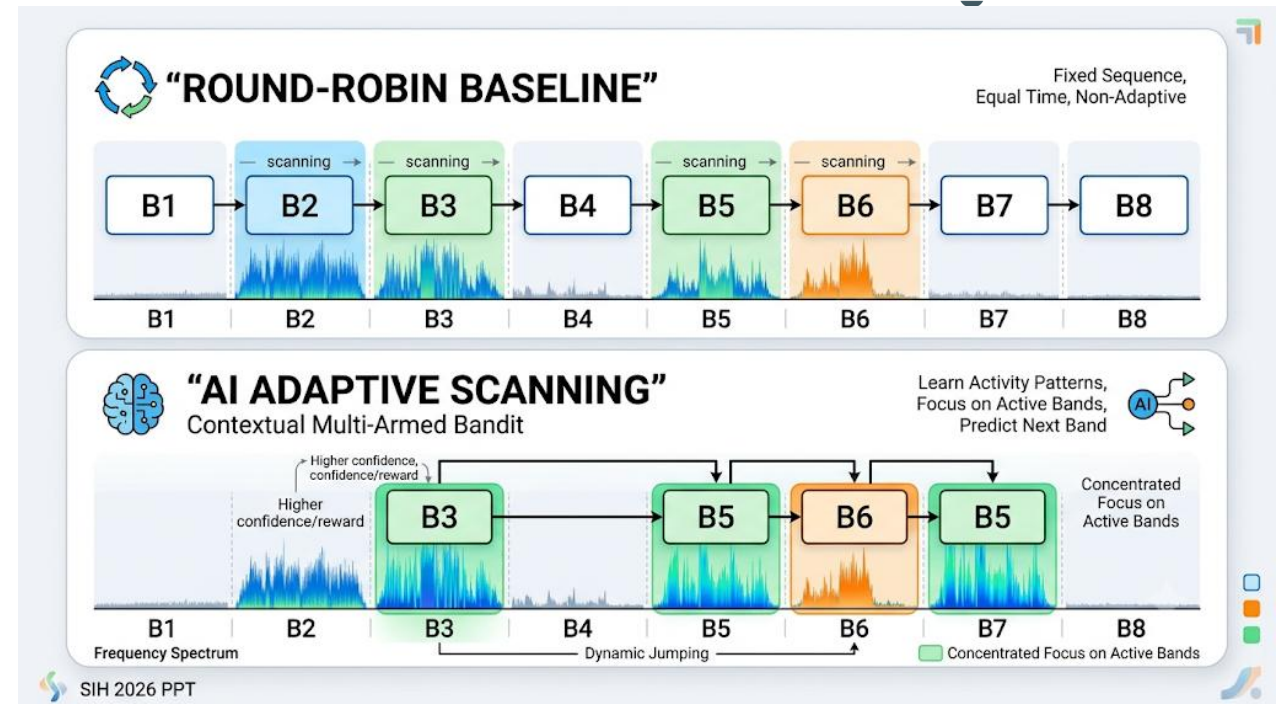
Add periodicity estimator to predict next-active windows.

Train on hits and misses through a Detection → Reward → Learning loop on every step.

Replay the Turing synthetic radar dataset — the dataset named in the problem statement.

Compare Baseline vs ML on the figures of merit the problem statement names.

Verified: 167 + 77 tests · 87% / 96% coverage · decision p95 6.6 ms



Feasibility Analysis

Simulation-first: no physical RF hardware; the system runs in a synthetic environment.

Already running: four services, one command each, Dockerised, on H2 or Postgres.

Verified: 167 scheduler tests and 77 estimator tests, at 87% and 96% coverage.

Reproducible: fixed seeds give every policy an identical spectrum.

Lightweight ML: the bandit needs no pre-training; DQN and PPO sit behind a flag.

Potential Challenges and Risks

ML does not win every metric — at the default weight it catches fewer distinct bursts than the sweep, 0.174 against 0.413.

Raw average intercept time ranks the better policy worse, because it averages only the bursts caught.

Periodicity does not yet help end to end — it is never told when a band was heard and found silent.

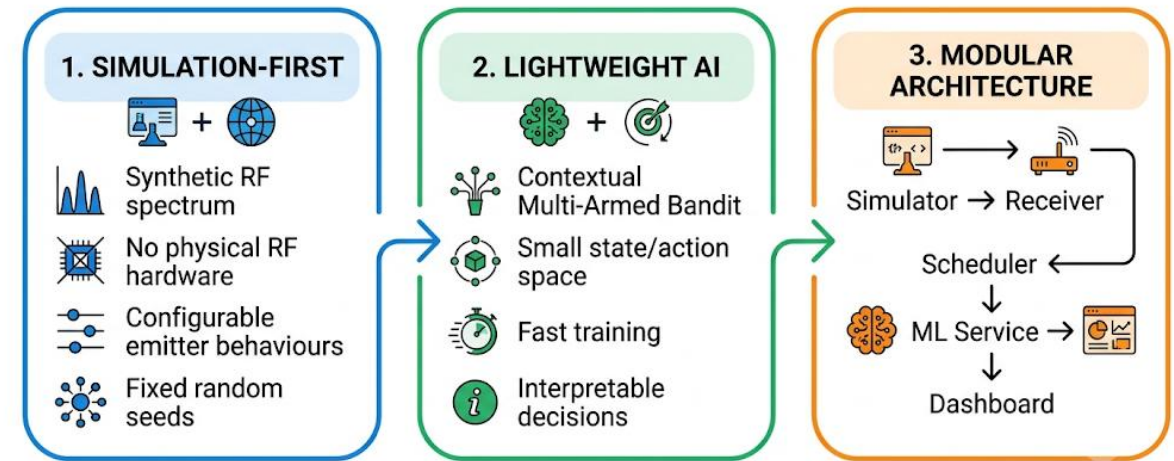
Strategies to Overcome Challenges

One reward weight slides along it. At **w5 = 20**, **burst coverage reaches 0.423** and **censored intercept time 577**, **beating the sweep's 588** — the mission picks the operating point.

Report **censored intercept time** beside the raw figure, charging undetected bursts the full episode.

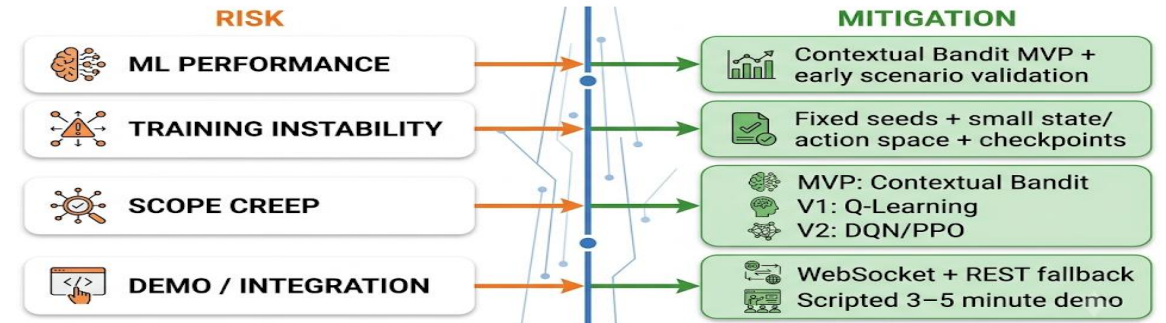
Named fix: carry **scanned-but-empty observations** in the contract. The test stays in the suite, disabled with the reason.

WHY THE SOLUTION IS FEASIBLE



Hackathon-Ready • Reproducible • Extensible

SIH 2026 - RISK MITIGATION STRATEGY



START SIMPLE → VALIDATE EARLY → SCALE GRADUALLY

For Spectrum Monitoring:

Listening efficiency rises **0.198 → 0.573** — over half of dwell now lands on live air instead of one fifth.

Smarter allocation for a receiver that hears only 2 of 16 bands at any instant.

For Detection Performance:

Probability of detection rises **0.107 → 0.345**, a $3.2\times$ gain on identical spectra.

High-priority detection rises **0.101 → 0.375**, a $3.7\times$ gain.

Average intercept time falls **2.34 → 0.59 steps** on the bursts we catch.

False alarms are **unchanged at 0.070** against 0.068 — the gain is not bought with false alarms.

For Research & Evaluation:

Objective Baseline versus ML comparison across 7 scenarios with fixed seeds.

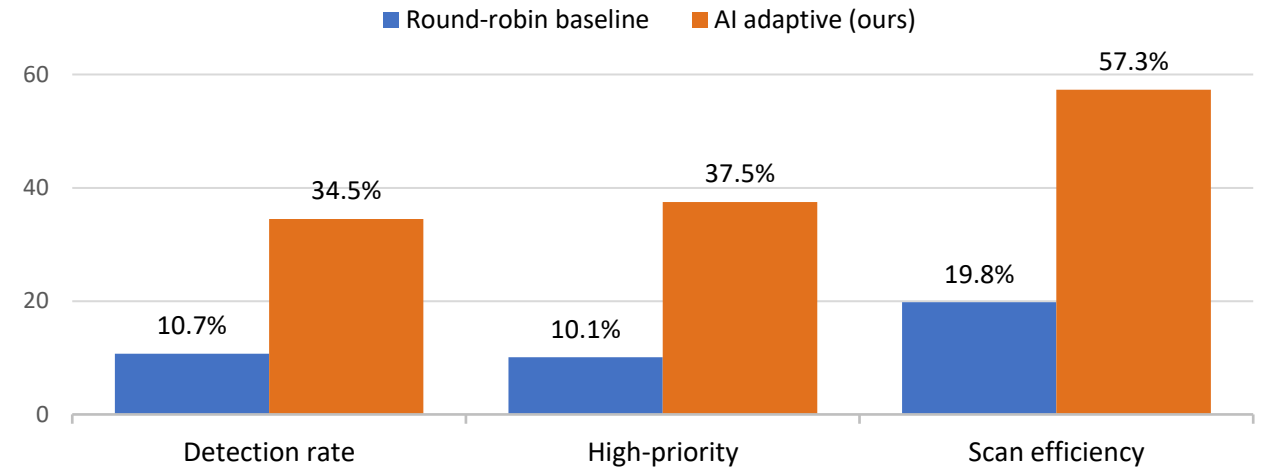
One script reproduces every number on this slide from a clean checkout.

For Future Development:

Modular architecture supports new emitter models and receiver constraints.

Q-Learning, DQN and PPO are already built and verified behind a flag.

ROUND-ROBIN SWEEP vs AI ADAPTIVE SCANNING (%)



FIGURES OF MERIT — SCENARIO B, 5 SEEDS, 1000 STEPS

Figure of merit	Random	Round-robin	Ours
Probability of detection (Pd)	0.104	0.107	0.345
Probability of false alarm (Pfa)	0.069	0.068	0.070
High-priority detection	0.109	0.101	0.375
Scan efficiency	0.190	0.198	0.573
Avg intercept time, caught ↓	2.46	2.34	0.59
Avg intercept time, censored ↓	660.7	587.8	826.1
Distinct bursts caught	0.340	0.413	0.174

↓ lower is better. Green marks the two rows the sweep still wins at our default reward weight — we report them rather than dropping them.

Research Papers

1. Gunn et al. — The Turing Synthetic Radar Dataset (*the dataset named in the problem statement*)
turing.ac.uk/news/publications/turing-synthetic-radar-dataset
2. Hashmi et al. — Artificial Intelligence Meets Radar Resource Management
doi.org/10.1049/rsn2.12337
3. Lu et al. — Constrained Deep Reinforcement Learning for Cognitive Radar Resource Management

Prior Art

US 10,523,342 B1 — Autonomous reinforcement learning method of receiver scan schedule control. BAE Systems.

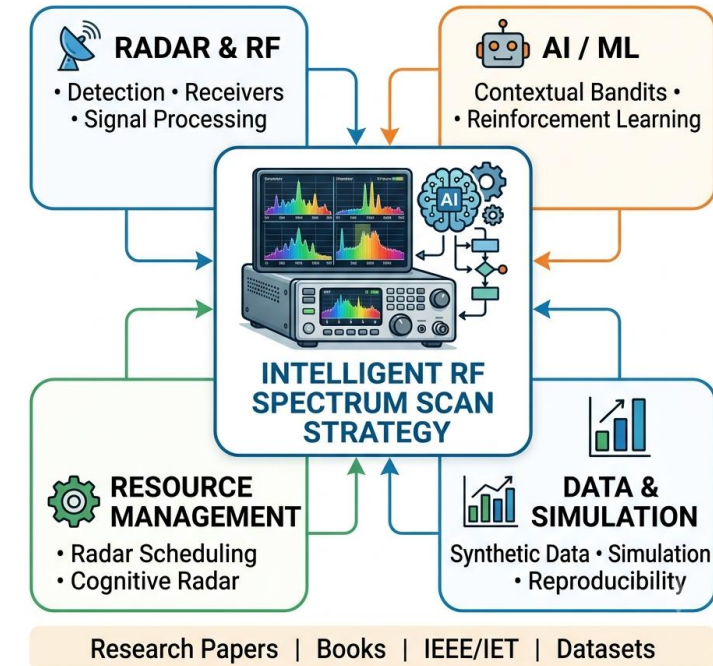
Method

1. Sutton and Barto — Reinforcement Learning: An Introduction, 2nd ed. MIT Press, 2018.
2. Auer, Cesa-Bianchi and Fischer — Finite-time analysis of the multiarmed bandit problem. Machine Learning 47, 2002.
3. Mardia and Jupp — Directional Statistics. Wiley, 2000 (Rayleigh test).

Books

1. Paul Hannen — Radar and Electronic Warfare Principles for the Non-Specialist.
2. Merrill I. Skolnik — Radar Handbook, 3rd edition.

RESEARCH AND REFERENCES



Prototype and source

Repository: github.com/aryan-stack-cloud/sih **Demo video:** [[add your link](#)]

scripts/compare.py reproduces every number in this deck from a clean checkout.

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